VEETRACER

Product & System Design Document

GPS tracking SaaS — Web, Mobile and Parent apps



Version	2.0 (Approved)
Date	18 October 2026
Status	Approved — development baseline
Scope	Product requirements, architecture, data model, roadmap

1 About this document

This document describes **VeeTracer**, a multi-tenant GPS tracking SaaS platform for the Indian market, covering the web dashboard, the VeeTracer mobile app and the VeeTracer Parent app. It consolidates every requirement agreed during the discovery discussions and turns them into a product scope, a system design, a data model and a phased roadmap.

Status: Approved on 18 October 2026 as the baseline for development. Any change after approval is recorded as a new version in the change table below.

1.1 How to read and request changes

- Every feature has a unique ID (for example **MAP-07**, **ALR-12**). Refer to IDs when asking for changes.
- Every feature has a phase: P1 core launch, P2 operations, P3 video & advanced, P4 growth. Later-phase features are still accounted for in the data model from day one, so adding them later needs no rework.
- Section 22 lists the decisions that are still open. Those need an answer before or during Phase 1.

1.2 Changes in this version

Version	Change
2.0	**Approved baseline.** Added the milestone delivery plan (section on phased roadmap). Feature scope unchanged from 1.7.
1.7	Web UI built with Tailwind CSS (+ shadcn/ui components); light, dark and system themes across web, mobile apps and maps; shared design tokens (module UI, section 11.2).
1.6	App line-up: separate VeeTracer Admin app; one role-based VeeTracer app for drivers, employees, users, individuals and delivery staff; Driver and Employee experiences; animations and trip playback across web and mobile (new section 11; modules APP, DRA, EMA, ANI).
1.5	Template Master page listing all templates (TPL-16 to TPL-21, section 9.5). New section 10: Master pages — a standard list/view/edit page for every master, the complete master catalogue and lookup masters (module MST).
1.4	Viewable and editable message templates (email, SMS, WhatsApp, push) with placeholders, preview, test send, versions, and the DLT/Meta approval workflow (module TPL, section 9.5).
1.3	Recipient management per alert: email, SMS, WhatsApp and push lists; company default lists; use default / default + extra / replace; external contacts; test send; delivery log (ALM-28 to ALM-38, section 9.4).
1.2	Configurable overspeed and long-idling alerts: thresholds, durations, tolerance, severity tiers, time/zone-based limits, repeat behaviour, exclusions (features ALM-18 to ALM-27, section 9.3).
1.1	Added Alert management (group level and individual level): configuration levels with inheritance and overrides, alert templates, user subscriptions, contact groups, snooze/maintenance mode, alert handling workflow and analytics (module ALM, section 9).
1.0	First complete draft.

1.3 Document contents

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2 Executive summary

VeeTracer lets businesses and individuals track anything that moves — vehicles, people, delivery staff, students' buses, assets, pets — on a live map, receive configurable alerts within seconds, replay history, manage drivers, routes and trips, monitor fuel and driving behaviour, and (later) watch live camera feeds. It is sold directly to customers on fixed monthly or annual plans in INR, and can be white-labelled and resold by partners.

2.1 Key decisions

Area	Decision
Market	India first. INR pricing, GST invoicing, Razorpay, DLT-registered SMS, AIS-140 support.
Sales model	Direct to customers (Business and Individual accounts). White-label partners later (P4).
Scale target	~2,000 devices in year one; architecture designed and load-tested for **10,000+**.
Top priority	Performance of **live tracking and alerts**: device → map under 2 s, alert → app under 2 s.
Devices	Mostly Concox (GT06 family); other brands, OBD, GPS watches, pet trackers, phones, AIS-140, dashcams/MDVR.
Configuration	Everything (alerts, map fields, reports, commands) driven by the **device model's capabilities**.
Objects vs devices	A vehicle/person/pet is permanent; the device fitted to it can be swapped without losing history.
Platform owner	Exactly **one Super Admin** (VeeTracer) who sees and controls every customer.
Organisation	Company → Branch → Vehicles; roles Admin → Manager → User; menu-level View/Create/Edit/Delete.
Data retention	Per customer, 30–180 days (6 months max), automatic deletion; no data loss under load.
Maps	OpenStreetMap data hosted locally: own tiles, address lookup and routing.
Technology	Open source first: NestJS, PostgreSQL + TimescaleDB, Valkey, React, Flutter, ZLMediaKit.
Hosting	Own server or any cloud, packaged with Docker.

2.2 Applications

Application	Platform	Used by
VeeTracer Web	Browser (desktop, tablet)	Super Admin, partners, all company roles, individuals — full management
VeeTracer Admin (mobile)	Android, iOS (Flutter)	Company Admins and Managers (and Super Admin) — fleet management on the go
VeeTracer (mobile, role-based)	Android, iOS (Flutter)	Drivers, attendants, employees (passengers), users/viewers, individuals, delivery staff — menus change by role
VeeTracer Parent	Android, iOS (Flutter)	Parents only — their own children's bus and trips
Public API & webhooks	REST / WebSocket	Customers' and partners' own systems

3 Product scope

3.1 What can be tracked

Object type	Typical device	Typical customer
Vehicle (car, bike, truck, bus, van)	Concox GT06 family, Teltonika, OBD, AIS-140, dashcam/MDVR	Fleets, logistics, schools, rentals, individuals
Person / employee	VeeTracer app on phone, GPS watch	Field staff, security, elderly care
Delivery person	VeeTracer app on phone (tracker mode)	Delivery and courier businesses
Student (via the bus)	Bus tracker + RFID reader (optional)	Schools
Asset	Battery tracker (GPS + cell tower/Wi-Fi)	Equipment, containers, generators
Pet	Pet tracker	Individuals

3.2 Account types

Capability	Business account	Individual account
Live map, history, alerts, geofences	Full	Full (simplified screens)
Automatic trips	Yes	Yes ("My rides")
Branches, roles, managers	Yes	No — single owner, can invite family as viewers
Drivers, licences, Driver → Vehicle → Route → Trip	Yes	No — the owner rides their own vehicles
Passenger module (students / employees)	Yes (module switch)	No
Fuel, driver score, maintenance	Yes	Basic (service & insurance reminders)
Home screen	Fleet dashboard	"My Vehicles"

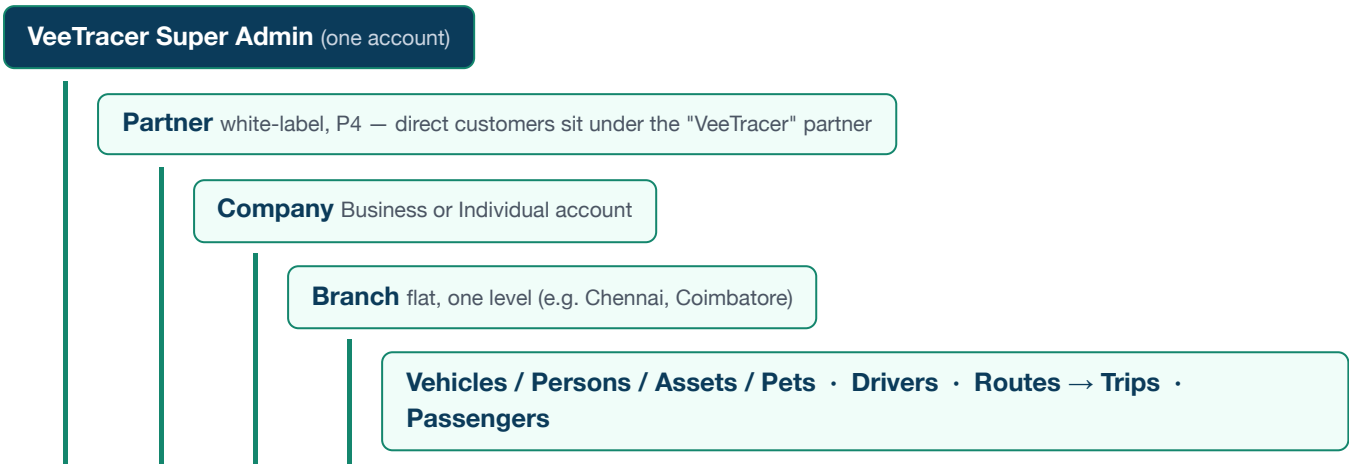
The account type is chosen at signup and controls menus and available plans. An Individual account can be upgraded to Business at any time without losing data.

3.3 Who uses the platform

Level	Role	Sees
Platform	**Super Admin** (exactly one — VeeTracer)	Everything: all partners, companies, users, devices, billing, settings
Partner (P4)	Partner Admin	Only its own customers (white-label reseller)
Company	Admin	Whole company
Company	Manager	Assigned branches and own team
Company	User / Viewer	Assigned branches and/or individual vehicles
Company	Driver / Attendant	Own duty, own vehicle, trip, attendance marking
Company	Passenger (Employee)	Own cab only ("Track my cab")
Company	Parent (Parent app)	Own children's details, bus and trips only

4 Organisation, roles and access

4.1 Hierarchy



4.2 Two independent access controls

Control	Question it answers	Configured by
Permissions (role → menu → action)	What can this user *do*?	Roles, menu-role mapping, user-role mapping
Data scope (branches + individual objects)	Which objects can this user *see*?	User → branch mapping, user → object mapping

- A vehicle belongs to **exactly one branch**.
- A user can be mapped to one or more branches **and** to individual objects. Example: one user sees their bike and their car on their own dashboard even if they sit in different branches.
- Saved **vehicle groups** (e.g. "Night shift trucks") are only for filtering and reports; they never grant visibility.
- Every API call, live update, report, export and webhook applies both controls on the server side.

4.3 Default permission matrix

Actions per menu: **V** view, **C** create, **E** edit, **D** delete, plus special actions (Export, Send command, View live video, Download video). Companies can create custom roles.

Menu	Admin	Manager	User	Viewer
Dashboard / Live map	V	V	V	V
Vehicles & objects	V C E D	V E	V	V
Devices & SIMs	V C E D	V	—	—
Drivers	V C E D	V C E	V	—
Routes & trips	V C E D	V C E	V	V
Passengers	V C E D	V C E	V	—
Geofences & POIs	V C E D	V C E	V	V
Alert rules	V C E D	V C E	—	—

Menu	Admin	Manager	User	Viewer
Alerts	V + acknowledge	V + acknowledge	V + acknowledge	V
Commands (engine cut-off etc.)	Send	Send	—	—
Live video / playback / download	All	View, playback	—	—
Reports & scheduled reports	V + Export + schedule	V + Export + schedule	V	V
Users & roles	V C E D	V C (own team)	—	—
Message templates	V E	V	—	—
Company settings, billing	V E (Owner pays)	—	—	—

4.4 Single Super Admin

- Exactly one Super Admin account exists. The system refuses to create a second one.
- Mandatory two-factor login. All Super Admin actions are written to the audit log.
- A documented server-level recovery procedure covers loss of the password or second factor.
- Super Admin can view any customer account ("view as") for support; every such session is logged.

5 Complete feature list

All features agreed so far, grouped by module. The count is computed automatically at the end of this section.

5.1 Platform, accounts and white-label

ID	Feature	Phase
PLT-01	Hierarchy Super Admin → Partner → Company → Branch → Objects (partner level enabled in P4)	P1
PLT-02	Complete data isolation between partners and companies	P1
PLT-03	Company profile: logo, address, GSTIN, timezone, units (km/mi, L/gal), language	P1
PLT-04	Account types Business and Individual, each with its own menus, plans and home screen	P1
PLT-05	Upgrade Individual → Business without losing data	P1
PLT-06	Customer self-signup and device activation by entering or scanning the IMEI	P1
PLT-07	Module switches per company (e.g. Passenger module, Video, Fuel) controlled by Super Admin and plan	P1

ID	Feature	Phase
WL-01	Partner custom domain, logo, colours, favicon and login page	P4
WL-02	Partner-branded email/SMS templates and sender identity	P4
WL-03	Partner-branded mobile and parent apps (separate store builds from one codebase)	P4
WL-04	Partner admin panel: own customers, devices, plans, prices	P4
WL-05	Partner plans on top of VeeTracer wholesale price per device	P4
WL-06	Two-level billing: VeeTracer invoices partner, partner invoices customers	P4
WL-07	Optional "Powered by VeeTracer"	P4

5.2 Login and security

ID	Feature	Phase
AUTH-01	Login with mobile number + OTP, or email + password	P1
AUTH-02	OTP: 6 digits, 5-minute validity, resend after 30 s, max 5 attempts, SMS (DLT) or WhatsApp	P1
AUTH-03	Password reset by email: single-use link valid 30 minutes; all sessions signed out after reset	P1
AUTH-04	Password policy and temporary lockout after repeated failures	P1
AUTH-05	Parent app: mobile + OTP only, matched to guardian phone in student profile	P1
AUTH-06	Two-factor login (mandatory for Super Admin, optional for others)	P1
AUTH-07	Trusted devices, active-session list and remote sign-out	P2
AUTH-08	Biometric unlock in mobile apps	P2

ID	Feature	Phase
SEC-01	Audit log of every create/update/delete, command, video view and login	P1
SEC-02	HTTPS/TLS everywhere; encrypted backups and encrypted sensitive documents	P1
SEC-03	DPDP Act 2023: consent records for person tracking, data export and deletion on request	P1
SEC-04	Aadhaar: only last 4 digits shown, document stored encrypted	P1
SEC-05	Rate limiting and abuse protection on login, OTP, API and ingestion	P1

5.3 Organisation, users and roles

ID	Feature	Phase
ORG-01	Branches (single level) under a company; each vehicle in exactly one branch	P1
ORG-02	Hierarchy Admin → Manager → User; managers manage only their own team and branches	P1
ORG-03	User → branch mapping and user → individual vehicle mapping	P1
ORG-04	Built-in roles plus custom roles per company	P1
ORG-05	Menu-role mapping with View/Create/Edit/Delete and special actions	P1
ORG-06	Menus on web and mobile shown or hidden from permissions	P1
ORG-07	Limited roles: Driver, Attendant, Passenger (Employee), Parent	P1
ORG-08	Saved vehicle groups for filters and reports (no effect on visibility)	P1
ORG-09	Super Admin "view as customer" with audit trail	P2

5.4 Devices and tracked objects

ID	Feature	Phase
DEV-01	Object types: Vehicle, Person, Delivery person, Asset, Pet, Watch wearer	P1
DEV-02	Object separated from device; device swap keeps history, rules, odometer and engine hours	P1
DEV-03	Device assignment history (which device on which object, from–to)	P1
DEV-04	Device model catalogue: brand, model, protocol, capabilities, extra fields, commands, time offset	P1
DEV-05	Capability-driven UI: alerts, map popup, reports and commands show only what the model supports (e.g. AC on/off)	P1
DEV-06	When a swapped-in device lacks a capability, dependent rules pause and are flagged (not deleted)	P1
DEV-07	Common fields + model-specific extra fields + company custom fields	P1
DEV-08	Object fields per type (vehicle: plate, make, fuel type; person: phone, employee ID; pet: breed, vet)	P1
DEV-09	Custom icons and colours per object	P1
DEV-10	Device & SIM stock: IMEI, SIM number, operator, expiry, assigned customer	P1
DEV-11	Device health: last seen, GSM signal, battery, firmware, data usage	P2
DEV-12	Bulk import of devices and objects from Excel	P2

5.5 Device protocols and hardware support

ID	Feature	Phase
PRO–01	Concox GT06 family (own decoder): login, GPS, LBS, status, alarm, heartbeat, commands	P1
PRO–02	VeeTracer app phone tracking; Traccar Client / OsmAnd HTTP protocol	P1
PRO–03	Normalisation of vendor codes into standard fields (e.g. Teltonika IO 239 → ignition)	P1
PRO–04	AIS-140 VLTD: packet decoder, emergency/panic handling, certificate records, government server forwarding	P2
PRO–05	Teltonika Codec 8 / 8E	P2
PRO–06	Other brands via Traccar gateway (Queclink, Meitrack, Coban, Sinotrack, Ruptela, 200+)	P2
PRO–07	OBD devices: RPM, coolant temp, fuel %, engine load, battery voltage, fault codes (DTC), VIN, odometer	P2
PRO–08	GPS watches: SOS, fall, heart rate, steps, watch removed, Wi-Fi/LBS location	P2
PRO–09	Pet trackers: safe zone, lost mode, battery-saving schedules	P2
PRO–10	Cell-tower and Wi-Fi location lookup (OpenCelliD + local Wi-Fi database)	P2
PRO–11	JT/T 808 dashcams and MDVRs (positions, alarms, ADAS/DMS events)	P3

5.6 Live tracking and map

ID	Feature	Phase
MAP–01	Live map with device → screen latency under 2 seconds	P1
MAP–02	Movement status colours: Running, Idle, Off, Overspeeding, Offline; thresholds per company	P1
MAP–03	View modes: all, by branch, by vehicle group, by status, by type, multiple selected, single vehicle (follow)	P1
MAP–04	Search by name, plate, IMEI, driver name or phone	P1
MAP–05	Vehicle list: status colour, speed, time in status, last update, driver, ignition/battery/GPS icons, status-count filter chips	P1
MAP–06	Marker clustering, direction arrows, plate labels, smooth marker animation	P1
MAP–07	Only objects in the visible area and within the user's scope are streamed	P1
MAP–08	Auto-fit map to selection; follow mode	P1
MAP–09	Vehicle info panel with tabs: Overview, Device, Sensors, Driver, Trip, Alerts, Documents, Camera	P1
MAP–10	Quick actions: follow, playback, today's trips, share link, command, video, geofence here, nearest vehicles, report	P1
MAP–11	Multi-screen tracking: 2, 4 or 6 maps following different vehicles	P2
MAP–12	Saved map views (filters + area)	P2
MAP–13	Live location share link with expiry	P2
MAP–14	Nearest vehicle to a point; distance and ETA (OSRM)	P2
MAP–15	Map tools: address search, distance measure, draw geofence	P1

ID	Feature	Phase
MAP-16	Layers: geofences, POIs, route lines, alert markers, labels	P1
MAP-17	Satellite layer (optional, needs a paid imagery provider)	P4

Movement status rules (defaults, configurable per company)

Status	Colour	Rule
Running	Green	Speed > 5 km/h
Idle	Amber	Ignition on, not moving for > 2 min
Off / Stopped	Red	Ignition off, or stationary when the device has no ignition input
Overspeeding	Purple	Speed above the object's (or zone's) limit
Offline	Grey	No data for > 10 min

5.7 History and playback

ID	Feature	Phase
HIS-01	Route playback 1×–16× with timeline scrubber and speed graph	P1
HIS-02	Stops, idle points and parking durations on the route	P1
HIS-03	Automatic trip list (from ignition/movement): start/end, distance, duration, max/avg speed	P1
HIS-04	Alerts shown on the playback timeline	P1
HIS-05	History continuous across device swaps	P1
HIS-06	Snap route to roads (OSRM)	P2
HIS-07	Planned route overlaid on actual path for route trips	P1
HIS-08	Video synced with playback	P3

5.8 Alerts and notifications

ID	Feature	Phase
ALR-01	Rules engine: scope (objects/branch/group/model), conditions, thresholds, schedule, cooldown, recipients, channels	P1
ALR-02	Alert types offered only if the device model supports them	P1
ALR-03	Latency targets: in-app under 2 s, push under 5 s	P1
ALR-04	Channels: in-app, push (FCM), email	P1
ALR-05	Channels: SMS (DLT), WhatsApp, webhook	P2
ALR-06	Live alert panel beside the map with sound, priority colours, filters, unread count	P1
ALR-07	Click alert → zoom to location and open the object; acknowledge/comment in place	P1
ALR-08	Critical alerts (SOS, fuel theft, crash, student not dropped) stay on screen until acknowledged	P1
ALR-09	Alert markers on the map	P1

ID	Feature	Phase
ALR-10	Acknowledge, comment, resolve; full alert history	P1
ALR-11	Escalation if unacknowledged after X minutes	P2
ALR-12	Per-user notification preferences and quiet hours	P2
ALR-13	Plan controls which channels are available (e.g. SMS/WhatsApp only on paid plans)	P1

Alert catalogue

Category	Alerts	Requires capability	Phase
Driving	Overspeed, idling, after-hours use, unauthorised movement	GPS (+ ignition for idling)	P1
Driving	Harsh braking / acceleration / cornering, towing	Accelerometer events or 1 s data	P2
Location	Geofence enter/exit, stayed too long, zone speed limit, no-go, keep-inside	GPS	P1
Location	Route deviation, POI arrival	GPS + route	P2
Device	Power cut, low battery, offline, GPS lost, tamper	Per model	P1
Device	SIM change	Per model	P2
Vehicle	Ignition on/off, AC on/off	Ignition / AC input	P1
Vehicle	Door, temperature, engine fault (DTC)	Sensor / OBD	P2
Fuel	Fuel theft/drain, refuel, low fuel	Fuel sensor or OBD	P2
Safety	SOS / panic	SOS button	P1
Safety	Crash, fall, watch removed	Per model	P2
Trips	Trip not started, late at stop, skipped stop, unscheduled stop, long halt, trip completed	Route trips	P1
Passengers	Student not dropped, wrong bus, absent, night-drop rules	Passenger module	P2
Maintenance	Service due, document expiry, licence/badge expiry	—	P2
Dashcam AI	Drowsiness, distraction, phone use, smoking, forward collision, lane departure	ADAS/DMS camera	P3

5.9 Alert management (group and individual level)

ID	Feature	Phase
ALM-01	Alert settings at four levels: Company default → Branch → Vehicle group → Individual vehicle/person	P1
ALM-02	Most specific level wins, per alert type and per setting (on/off, threshold, schedule, channels, recipients, priority, cooldown)	P1
ALM-03	Every setting shows where it comes from (e.g. "inherited from Branch: Chennai") with one-click "reset to inherited"	P1
ALM-04	Individual override: different threshold or recipients for one vehicle, or switch an alert off for that vehicle only	P1

ID	Feature	Phase
ALM-05	Vehicle in several groups: admin sets group priority; conflicts are shown in the UI before saving	P1
ALM-06	Only alert types the object's device supports are shown at individual level; group level shows the union with a 'supported by N of M' count	P1
ALM-07	Alert templates (e.g. School bus pack, Truck pack, Two-wheeler pack, Pet pack) applied to a group or selected objects in one step	P1
ALM-08	Copy alert settings from one vehicle to other vehicles or groups	P1
ALM-09	Bulk enable/disable an alert type across a branch or group	P1
ALM-10	Snooze / maintenance mode: mute all or chosen alerts for an object or group until a date/time (e.g. vehicle in workshop); auto-resume	P1
ALM-11	Mandatory alerts (SOS, panic, crash, student not dropped) cannot be switched off below company level	P1
ALM-12	User subscriptions: each user chooses which alert types, vehicles/groups and channels they receive, within what the admin allows	P1
ALM-13	Contact groups for recipients, including people without a login (e.g. customer contact, security desk) — email/SMS/WhatsApp only	P2
ALM-14	Alert handling workflow: New → Acknowledged → In progress → Resolved / False alarm, with assignee, notes and resolution time	P2
ALM-15	Assign an alert to a user; escalation chain per level (user → manager → admin)	P2
ALM-16	Alert analytics: counts by type, group, vehicle, driver; response and resolution times; noisiest vehicles and rules	P2
ALM-17	Change history of alert settings (who changed what, at which level)	P1
ALM-18	Overspeed: speed limit per level (company, branch, group, vehicle) with vehicle-type defaults	P1
ALM-19	Overspeed: tolerance, minimum duration before alerting, Warning/Critical tiers	P1
ALM-20	Overspeed: time-based limits (e.g. lower limit at night) and zone limits inside geofences	P1
ALM-21	Overspeed: one alert per event, optional repeat while it continues, end-of-event summary (max speed, duration, distance)	P1
ALM-22	Overspeed: road speed limits from OSM maxspeed data where available	P4
ALM-23	Long idling: idle threshold minutes per level, Warning/Critical tiers, repeat interval while idling continues	P1
ALM-24	Long idling: idle detection by ignition, or by external voltage / OBD RPM for devices without ignition wire	P1
ALM-25	Long idling: tolerance so small creeping movement doesn't reset the timer	P1
ALM-26	Long idling: exclusions by geofence (e.g. loading bay) and by time window; separate 'idling with AC on' alert	P1
ALM-27	Long idling: end-of-idle summary with total duration and estimated fuel wasted	P2
ALM-28	Company default recipient lists: default email list, default SMS numbers, default WhatsApp numbers, default push users	P1
ALM-29	Per alert type and per level: separate Email list, SMS list, WhatsApp list and Push list	P1
ALM-30	List mode per channel: Use default list / Default + additional / Only this list / None	P1

ID	Feature	Phase
ALM-31	Recipients can be users, roles (e.g. all Branch Managers), dynamic contacts (assigned users, current driver, branch manager), contact groups or typed emails/numbers	P1
ALM-32	Branch and group default lists, inherited like other alert settings	P1
ALM-33	Severity routing: e.g. Warning → email only, Critical → email + SMS + WhatsApp	P1
ALM-34	Validation: email format, Indian mobile (+91, 10 digits), duplicates removed; optional OTP verification for external numbers	P1
ALM-35	'Send test' button per alert to check every list	P1
ALM-36	Per-recipient delivery log: sent, delivered, failed, with retry	P1
ALM-37	Unsubscribe link in emails and opt-out for external SMS numbers (Critical/SOS can be marked non-optional for staff)	P2
ALM-38	Plan limits on SMS/WhatsApp recipients per alert and per month; usage shown before saving	P1

5.10 Message templates (email, SMS, WhatsApp, push)

ID	Feature	Phase
TPL-01	Template library: view every email, SMS, WhatsApp and push template, filtered by alert type, channel, language, level	P1
TPL-02	Templates for every alert type plus system messages: OTP, password reset, welcome, report emails, invoices, renewal reminders, parent notifications	P1
TPL-03	Levels: system default (Super Admin) → partner (P4) → company override; 'reset to default' at any time	P1
TPL-04	Email editor: subject, rich-text/HTML body, company logo and colours, footer	P1
TPL-05	Placeholders per alert type, inserted from a list: {{vehicle_name}}, {{plate}}, {{alert_type}}, {{speed}}, {{limit}}, {{location}}, {{map_link}}, {{time}}, {{driver_name}}, {{geofence}}, {{duration}}…	P1
TPL-06	Live preview with sample or real data; send test to my email/mobile	P1
TPL-07	Validation: unknown placeholders blocked, required placeholders enforced	P1
TPL-08	SMS editor: character counter, parts count (160 GSM / 70 Unicode for Tamil/Hindi), cost estimate	P1
TPL-09	SMS DLT workflow: template text + DLT template ID + sender ID stored; only DLT-approved templates can be activated	P1
TPL-10	Companies choose among approved SMS templates per alert and map variables; new wording goes through an approval request	P1
TPL-11	WhatsApp templates with Meta approval status (pending/approved/rejected) synced from the provider	P2
TPL-12	Push notification title/body templates with sound and priority per alert type	P1
TPL-13	Version history with who/when, compare and restore	P1
TPL-14	Language variants per template (English first; Tamil/Hindi later)	P4
TPL-15	Permission 'Message templates' (View/Edit) in menu-role mapping	P1
TPL-16	Template Master page: one list of all templates (email, SMS, WhatsApp, push; alert and system messages)	P1

ID	Feature	Phase
TPL-17	Master columns: name, purpose/alert type, channel, level, language, status, DLT/Meta ID, used by, last changed by/at	P1
TPL-18	Master filters and search: channel, alert type, level, status, language, 'overridden by my company', 'not approved'	P1
TPL-19	Row actions: view, edit, preview, send test, duplicate, version history, reset to default, activate/deactivate	P1
TPL-20	Super Admin master shows system templates plus how many companies override each one, with drill-down	P1
TPL-21	Export the template list (Excel/CSV), e.g. for DLT registration	P1

5.11 Geofences and points of interest

ID	Feature	Phase
GEO-01	Shapes: circle, polygon, rectangle	P1
GEO-02	Route corridor (line + width) for deviation alerts	P2
GEO-03	Create by drawing on map (web and mobile)	P1
GEO-04	Create from address/place search with radius	P1
GEO-05	Create from a vehicle's current location or from my phone's location	P1
GEO-06	Create from a stop, POI or route (route stops become geofences automatically)	P1
GEO-07	Create from OSM boundaries: city, district, state, pincode area	P2
GEO-08	Bulk import from Excel/CSV, KML, GeoJSON	P2
GEO-09	Details: name, category (Home, Office, Depot, Customer, School, Parking, Fuel station, No-go, Safe zone), colour, icon, address	P1
GEO-10	Owned by a branch or shared company-wide	P1
GEO-11	Applies to all, a branch, a vehicle group or selected objects	P1
GEO-12	Checks: entry, exit, both; stayed longer than X min; zone speed limit; no-go; keep-inside	P1
GEO-13	Time windows (days and hours)	P1
GEO-14	Edit: drag points, move, resize; enable/disable; change history preserved for past reports	P1
GEO-15	List with search, filter and bulk actions	P1
GEO-16	Anti-flapping: boundary buffer and minimum dwell before enter/exit counts	P1
GEO-17	In-memory spatial index — checks in microseconds at 10,000+ devices	P1
GEO-18	Plan limits on number of geofences	P1
GEO-19	Points of interest with arrival detection	P2
GEO-20	Reports: visits (in/out/duration), entry/exit log, time per geofence per day, no-go violations	P2

5.12 Device commands

ID	Feature	Phase
CMD-01	Engine cut-off / restore, request location	P1
CMD-02	Change interval, reboot, set server/APN, custom commands	P2
CMD-03	Commands offered depend on the device model	P1
CMD-04	Status: sent → acknowledged / failed / expired; full audit log	P1
CMD-05	Safety: permission-based, confirmation dialog, blocked above a configurable speed	P1
CMD-06	Offline queue with expiry; SMS fallback through an SMS gateway	P2
CMD-07	Watch commands: call device, listen-in, two-way calling	P2

5.13 Video and cameras

ID	Feature	Phase
VID-01	Camera capabilities in device model: channels, live, playback, snapshot, two-way audio, protocol	P3
VID-02	Live view of 1–8 channels beside the map	P3
VID-03	On-demand streams with auto-stop to save SIM data and bandwidth	P3
VID-04	Snapshots, manual or automatic on alerts	P3
VID-05	Alarm video clips attached to alerts and timeline	P3
VID-06	Playback from device SD card, synced with route playback	P3
VID-07	Two-way talk with driver	P3
VID-08	Sources: JT/T 1078, RTMP, RTSP/ONVIF IP cameras, photo uploads	P3
VID-09	Separate permissions and audit log for live view, playback and download	P3
VID-10	Video retention and minute limits per plan	P3

5.14 Fuel monitoring

ID	Feature	Phase
FUEL-01	Sources: analog/digital fuel sensor, OBD fuel level, manual fill-up entries	P2
FUEL-02	Calibration table per tank; multiple tanks per vehicle	P2
FUEL-03	Smoothing of sloshing and slope effects	P2
FUEL-04	Refuel and theft/drain detection with time, litres and location	P2
FUEL-05	Consumption per trip/day, km/L, idle fuel, sensor vs manual comparison	P2

5.15 Drivers and driving behaviour (Business accounts)

ID	Feature	Phase
DRV-01	Driver profile: name, photo, phones, address, DOB, blood group, emergency contact, employee ID, joining date, branch	P1
DRV-02	Licence: number, vehicle classes (LMV, MCWG, HMV, Transport…), issue/expiry, RTO, document upload	P1
DRV-03	Commercial: transport/PSV badge and expiry, medical fitness, police verification status	P1
DRV-04	Optional ID proof upload (Aadhaar masked to last 4 digits, encrypted)	P1
DRV-05	Driver status: Active, On leave, Inactive (history kept)	P1
DRV-06	Driver → vehicle assignment with from–to times; one driver per vehicle at a time; full history	P1
DRV-07	Current driver shown on map, trips, alerts, reports	P1
DRV-08	Warning (or block, per company) when licence class doesn't cover the vehicle or licence is expired	P1
DRV-09	Licence/badge/medical expiry reminders at 30, 15, 7 days and on expiry	P2
DRV-10	Driver identification by RFID/iButton or app login	P2
DRV-11	Driver score 0–100 with configurable weights; rankings and trends	P2
DRV-12	Fatigue: continuous driving without a break	P2
DRV-13	Licence verification against Sarathi via authorised provider	P4

5.16 Routes and trips (Driver → Vehicle → Route → Trip)

ID	Feature	Phase
RTE-01	Route = ordered stops (start → stops → end) with arrival radius, planned arrival/departure, planned wait	P1
RTE-02	Route path drawn or generated along roads (OSRM), with corridor width	P2
RTE-03	Route schedule: days of week, start time	P1
RTE-04	Copy, edit, archive; editing never alters past trips	P1
RTE-05	Trip = Driver + Vehicle + Route + date/time; one-off or recurring (auto-generated)	P1
RTE-06	Trip states: Scheduled → Started → In progress → Completed / Incomplete / Cancelled	P1
RTE-07	Auto start on leaving the start point and auto complete at the end; manual override	P1
RTE-08	Per stop: planned vs actual arrival, in time, out time, dwell, status (on time/late/early/skipped)	P1
RTE-09	Unscheduled stops recorded with location and duration	P1
RTE-10	Live trip view: progress, current stop, ETA to next stop, delay	P1
RTE-11	Trip alerts (see alert catalogue)	P1
RTE-12	Complete trip record: path, stops, distance, duration, speeds, idle, alerts, score, fuel, video	P1
RTE-13	Trip playback with planned route and stops over actual path	P1
RTE-14	Trip reports: on-time performance by route/driver/vehicle, stop delays, planned vs actual km	P1

ID	Feature	Phase
RTE-15	Routes optionally assignable to delivery persons	P2

5.17 Passenger management (students and employees)

ID	Feature	Phase
PAS-01	Student profile: name, photo, admission no., class/section, campus, DOB, blood group, medical notes, guardians, address, RFID	P2
PAS-02	Employee profile: name, photo, employee ID, department, designation, shift, phone, address, emergency contact, gender, RFID	P2
PAS-03	Bulk import; active/inactive with history	P2
PAS-04	Mapping passenger → vehicle, route, pickup stop and drop stop (separate morning/evening)	P2
PAS-05	Mapping history with effective dates; seat-capacity warnings	P2
PAS-06	Passenger list per trip and per stop	P2
PAS-07	Attendance optional per company: Off / RFID / in-app / both (merged, no duplicates)	P2
PAS-08	Boarding and drop records with time and location; wrong-vehicle detection	P2
PAS-09	Safety alerts: student not dropped, wrong bus, absent, night-drop rules for women employees	P2
PAS-10	VeeTracer Parent app: own children only — profile, bus live location in trip hours, ETA, boarding status, trip history	P2
PAS-11	Parent notifications: bus started, approaching stop, boarded, dropped, delayed	P2
PAS-12	Multiple children per parent login (even on different buses)	P2
PAS-13	Employee "Track my cab": live cab, ETA, boarding confirmation, SOS	P2
PAS-14	Reports: attendance, absentees, boarding logs, occupancy, notification log	P2
PAS-15	Module switch so it can be bundled or sold as an add-on later	P2

5.18 Mobile tracking for people and delivery staff

ID	Feature	Phase
MOB-01	Phone becomes the tracker; Start Duty / End Duty	P1
MOB-02	Background tracking (Android foreground service; iOS within its limits)	P1
MOB-03	Adaptive interval to save battery; offline buffering and later upload	P1
MOB-04	Person shown on map with photo, last update and phone battery	P1
MOB-05	Consent record; tracking only during duty	P1
MOB-06	Ad-hoc tracking: SMS/WhatsApp invite link to share location for a set time	P4

5.19 Maintenance and documents

ID	Feature	Phase
MNT-01	Service schedules by km, engine hours or date	P2
MNT-02	Vehicle documents with expiry reminders: insurance, permit, fitness, PUC	P2
MNT-03	Maintenance log and costs	P2

5.20 Reports

ID	Feature	Phase
RPT-01	Summary report per vehicle, day-wise / week-wise / month-wise: distance, running, idle, stop, offline time, trips, max/avg speed, overspeed count, first start, last stop	P1
RPT-02	Running + idle + stop + offline always add up to the full period; totals per vehicle and group	P1
RPT-03	Trips, stops, idle, alerts reports	P1
RPT-04	Geofence visits, ignition, overspeed, AC usage, engine hours, utilisation	P2
RPT-05	Fuel, driver score, attendance reports	P2
RPT-06	Export to Excel, PDF, CSV	P1
RPT-07	Scheduled reports: vehicles (branch / selection / vehicle group), report types, daily/weekly/monthly, time, format, recipients	P1
RPT-08	Scheduled report runs with the scheduler's current access; history, re-download, pause/resume	P1
RPT-09	Periods in company timezone; weeks start Monday (configurable); calendar months	P1
RPT-10	Pre-computed daily summaries so fleet-wide monthly reports open in seconds	P1
RPT-11	Charts: daily distance trend, running vs idle split	P1

5.21 Dashboards

ID	Feature	Phase
DSH-01	Company dashboard: status counts, mini live map, today's distance/running/idle, utilisation, alerts by type	P1
DSH-02	Widgets: fuel, driver ranking, maintenance due, documents expiring, devices not reporting, plan usage	P2
DSH-03	User-customisable widget layout	P2
DSH-04	Compact mobile dashboard; Individual "My Vehicles" home	P1
DSH-05	Super Admin dashboard: customers, devices online, data intake rate, revenue, renewals, failed payments, server health	P1
DSH-06	Partner dashboard	P4

5.22 Billing and subscriptions

ID	Feature	Phase
BIL-01	Fixed plans with monthly and annual prices in INR, separate for Business and Individual	P1

ID	Feature	Phase
BIL-02	Plan limits: devices, users, history days, geofences, channels, video minutes, modules	P1
BIL-03	Razorpay Subscriptions: auto-renew, grace period, auto-suspend, reactivation	P1
BIL-04	GST invoices (CGST/SGST/IGST, GSTIN), PDF and email	P1
BIL-05	SMS/WhatsApp credit packs	P2
BIL-06	Upgrade/downgrade with proration; coupons and discounts	P2

5.23 Integrations

ID	Feature	Phase
INT-01	REST API with per-company API keys and OpenAPI/Swagger documentation	P1
INT-02	Webhooks for positions, alerts, trips with signed payloads and retries	P2
INT-03	WebSocket live stream for customer systems	P2
INT-04	Raw data forwarding to other platforms, ERP or government servers (AIS-140)	P2
INT-05	Inbound positions from other GPS platforms	P2

5.24 Data management

ID	Feature	Phase
DATA-01	Retention per company 30/60/90/180 days (6 months max), automatic deletion, export before deletion	P1
DATA-02	Bad-point filtering: invalid fix, 0,0, low satellites/accuracy, impossible jumps (limit per object type), parked drift, duplicates, future timestamps	P1
DATA-03	Filtered points flagged, never deleted; raw packets logged for reprocessing	P1
DATA-04	Late/out-of-order points (sent after no-signal periods) placed correctly in history; summaries recalculated	P1
DATA-05	All storage in UTC; display in company/user timezone; device time offset per model	P1
DATA-06	Odometer and engine hours carried across device swaps	P1

5.25 Maps (self-hosted OpenStreetMap)

ID	Feature	Phase
OSM-01	India OSM extract → own vector tiles (Planetiler + Martin), displayed with MapLibre	P1
OSM-02	Reverse geocoding (address lookup) with Nominatim/Photon and caching	P1
OSM-03	Routing, ETA and road snapping with OSRM	P2
OSM-04	Monthly map data refresh	P2

5.26 Mobile apps

ID	Feature	Phase
APP-01	Three mobile apps from one Flutter codebase with shared modules: VeeTracer Admin, VeeTracer (role-based), VeeTracer Parent	P1
APP-02	VeeTracer Admin: dashboard, live map, list, vehicle tabs, history, trips, alerts + management (vehicles, drivers, routes, geofences, alert settings, users)	P1
APP-03	VeeTracer role-based app: after login the menus, home screen and permissions are built from the user's role and permissions	P1
APP-04	A user with several roles (e.g. driver who is also a viewer) can switch role inside the app	P2
APP-05	Push notifications with alert sounds and critical-alert handling (all apps)	P1
APP-06	Commands from mobile (Admin app, permission-based)	P1
APP-07	Live video in mobile	P3
APP-08	Offline mode: cached data, queued actions (attendance, fuel entries, inspections) synced when online	P2
APP-09	Tamil and Hindi languages	P4

5.27 Driver experience (VeeTracer app, Driver role)

ID	Feature	Phase
DRA-01	Login by mobile OTP; home shows today's duty, assigned vehicle and trips	P1
DRA-02	Start Duty / End Duty; phone tracking during duty if the vehicle has no device	P1
DRA-03	Current trip: route on map, ordered stop list with planned times, next stop ETA, turn-by-turn via Google Maps / OSM navigation app	P1
DRA-04	Automatic stop arrival/departure; manual 'Arrived' / 'Departed' / 'Skip with reason'	P1
DRA-05	SOS / panic button with location	P1
DRA-06	Passenger list per stop and attendance marking (driver or attendant role)	P2
DRA-07	Pre-trip vehicle inspection checklist with photos (tyres, lights, brakes, cleanliness)	P2
DRA-08	Fuel entry with receipt photo and odometer	P2
DRA-09	Own driving score, events and tips; overspeed voice warning while driving	P2
DRA-10	My documents: licence, badge, medical expiry reminders	P2
DRA-11	Trip history with animated replay	P1

5.28 Employee experience (VeeTracer app, Employee role)

ID	Feature	Phase
EMA-01	Login by mobile OTP; home shows today's pickup and drop, cab, driver photo and phone	P2
EMA-02	Track my cab live with ETA to my pickup point; 'cab approaching' notification	P2
EMA-03	Boarding confirmation (RFID tap, attendant marking, or confirm in app)	P2

ID	Feature	Phase
EMA-04	SOS button and share-my-ride link to family during the trip	P2
EMA-05	My shift and roster; request pickup point or shift change (approval by admin)	P2
EMA-06	Trip history and trip rating/feedback	P2

5.29 Super Admin console

ID	Feature	Phase
ADM-01	All partners, companies, users and devices; search across the platform	P1
ADM-02	Activate, suspend, change plan, change account type, enable modules	P1
ADM-03	Device model catalogue and capability management	P1
ADM-04	Device and SIM inventory, assignment to customers	P1
ADM-05	Plans and pricing management	P1
ADM-06	Platform settings: SMS/WhatsApp/email providers, templates, map services	P1
ADM-07	Revenue, renewals, failed payments, invoices	P1
ADM-08	System health: ingestion rate, queue depth, latency, connected devices, errors	P1

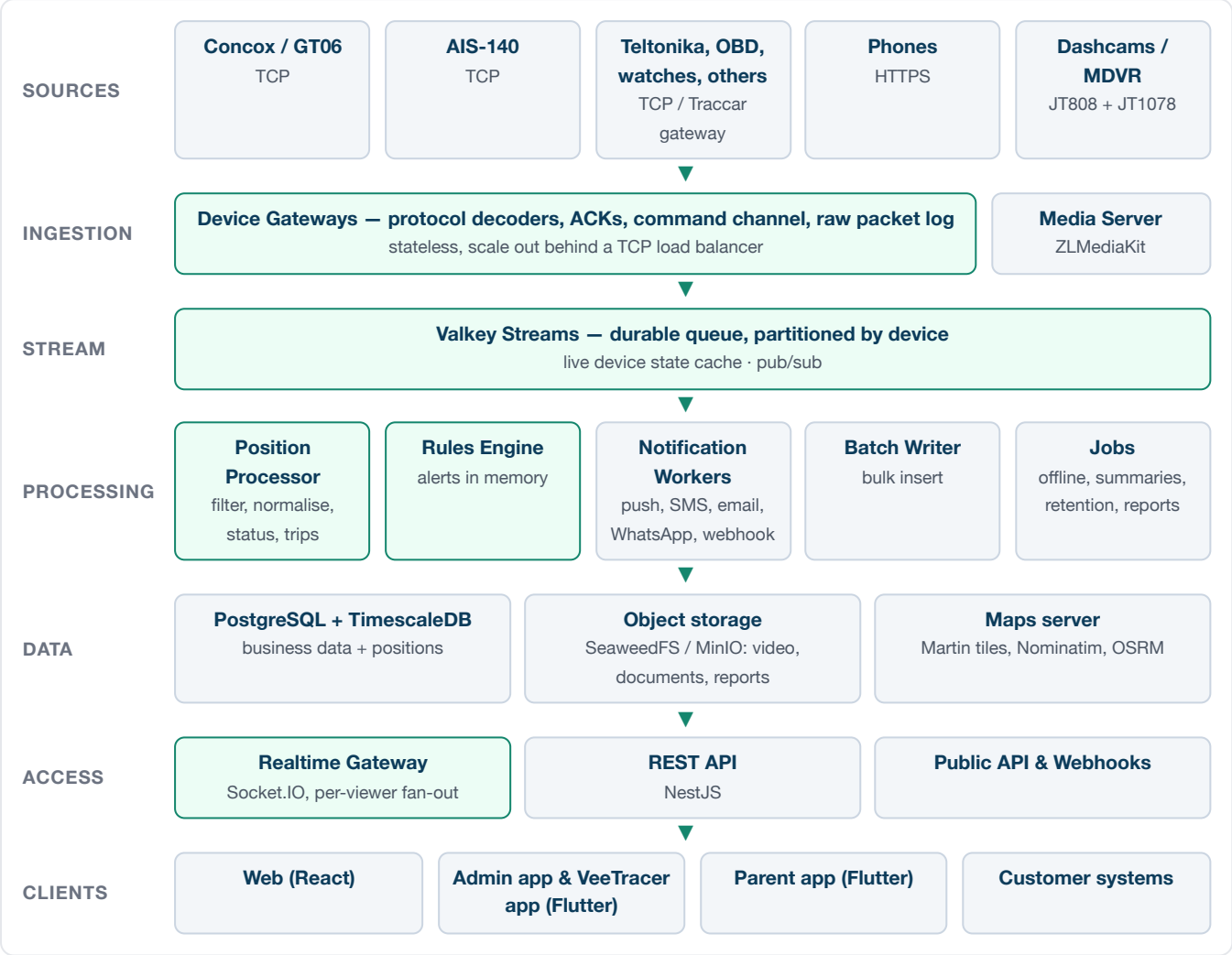
5.30 Performance and scalability

ID	Feature	Phase
PERF-01	Designed and load-tested for 10,000+ devices (~1,000 positions per second peak)	P1
PERF-02	Hot path (live + alerts) processed in memory, separated from storage and reporting	P1
PERF-03	Targets: device → map under 2 s, alert → in-app under 2 s, push under 5 s (95th percentile)	P1
PERF-04	Horizontally scalable gateways and workers	P1
PERF-05	TimescaleDB compression and partitioning	P1
PERF-06	Durable queue: no data lost during spikes or restarts	P1
PERF-07	Live updates batched per viewer (~1 s) and filtered by scope and viewport	P1
PERF-08	10,000-device simulator for load testing	P1
PERF-09	Monitoring and alerting (Prometheus + Grafana)	P1
PERF-10	Daily backups, point-in-time restore, disaster-recovery runbook	P1

Total: 369 features across 35 modules.

6 System architecture

6.1 Overview



☐ Hot path — optimised for live tracking and alert latency.

6.2 Life of a position (hot path)

Step	What happens	Budget
1	Device sends packet; gateway decodes, ACKs immediately, appends raw packet and normalised position to the stream	< 50 ms
2	Position processor reads the stream: bad-point filter, timezone, normalise attributes, update movement status, odometer, trip state	< 100 ms
3	Rules engine evaluates the object's rules against in-memory state (previous position, geofence membership via spatial index, cooldowns)	< 50 ms
4	Live state written to Valkey; update published to the realtime gateway; alerts published to notification workers	< 50 ms
5	Realtime gateway batches per viewer (≤ 1 s), filters by scope and viewport, pushes to web/mobile	≤ 1 s
6	In parallel (cold path): batch writer bulk-inserts positions into TimescaleDB; summaries and reports update later	async

> Database writes are never on the critical path for the live map or alerts. If the database slows down, the stream absorbs the backlog and nothing is lost.

6.3 Technology stack (open source first)

Layer	Choice	Licence / cost
Backend API, gateways, workers	Node.js + TypeScript (NestJS)	MIT — free
Database	PostgreSQL 16 + TimescaleDB (+ PostGIS for spatial queries)	Free (TimescaleDB community licence)
Cache, queues, pub/sub	Valkey (open-source Redis fork) + BullMQ	BSD / MIT — free
Realtime	Socket.IO (Valkey adapter for multiple nodes)	MIT — free
Web app	React + TypeScript + Vite, **Tailwind CSS** + shadcn/ui (Radix) components, MapLibre GL, Apache ECharts	MIT/Apache — free
Mobile apps	Flutter (maplibre, geolocator, background service)	BSD — free
Map tiles	OSM India extract → Planetiler → Martin	ODbL data, Apache/MIT tools — free
Geocoding / routing	Nominatim (or Photon), OSRM	GPL/Apache/BSD — free
Other device brands	Traccar (as protocol gateway)	Apache 2.0 — free
Video	ZLMediaKit (JT1078, RTMP, RTSP → WebRTC/HLS/FLV)	MIT — free
Object storage	SeaweedFS (or MinIO)	Apache 2.0 / AGPL — free
Monitoring	Prometheus, Grafana, Loki	Free
Push notifications	Firebase Cloud Messaging	Free (proprietary service)
Email	Amazon SES (or self-hosted Postal)	~$0.10 per 1,000 emails
Packaging	Docker Compose (single/multi server); Kubernetes optional later	Free

6.4 Scaling path

Stage	Devices	Deployment
Launch	up to ~3,000	2 servers: application server (gateways, API, workers, Valkey) + database server; maps on the app server or a third box
Growth	~10,000	2 gateway/API nodes behind a load balancer, dedicated worker node, database server with replica, maps server, media server (P3)
Beyond	50,000+	More gateway/worker nodes, stream partitions, TimescaleDB multi-node or read replicas, Kubernetes

7 Device integration

7.1 Device model catalogue

Every supported model is described once in the catalogue, maintained by the Super Admin. Everything the customer sees for a device is derived from it.

Catalogue field	Example (Concox GT06N)
Brand / model / protocol	Concox / GT06N / GT06
Capabilities	ignition, SOS, power-cut, engine cut-off, battery, LBS
Extra fields	external voltage (V), GSM signal (0–4), oil/electric relay state
Commands	engine cut-off, restore, locate, set interval (with SMS equivalents)
Time handling	UTC
Filtering defaults	max speed 200 km/h, parked-drift radius 30 m
Default alert templates	power cut, SOS, ignition on/off, overspeed

7.2 Capabilities

ignition · ac · door · fuel · temperature · sos · power_cut · battery · engine_cut · harsh_events · crash · obd · dtc · rfid · camera(channels) · adas · dms · two_way_audio · heart_rate · steps · fall · wear_detect · lbs · wifi_location · seatbelt

7.3 Normalised fields

Group	Fields
Core (every position)	fix time, server time, latitude, longitude, speed, course, altitude, satellites, accuracy/HDOP, valid fix, location source (GPS/LBS/Wi-Fi)
Common status	ignition, motion, battery %, external voltage, GSM signal, odometer, engine hours
Model-specific (attributes)	ac, door, fuel level/litres, temperatures, RPM, coolant temp, engine load, DTC list, heart rate, steps, RFID tag, ADAS/DMS events…

Each protocol decoder maps vendor codes into these names, so the same alert, report or map field works for any brand.

7.4 Protocol plan

Protocol / source	Approach	Phase
Concox GT06 family	Native decoder in our gateway	P1
Phone (VeeTracer app), OsmAnd/Traccar Client	HTTPS ingestion API	P1
AIS-140	Native decoder; forwarding to state backend as required	P2
Teltonika	Native decoder	P2
200+ other brands	Traccar gateway forwarding normalised positions to our stream	P2
JT/T 808 + 1078 (dashcams, MDVR)	Native JT808 decoder + ZLMediaKit for video	P3

7.5 Device swap

- Positions, trips, alerts and summaries are stored against the **object** (and the device that produced them).
- Swapping closes the old assignment and opens a new one; odometer and engine hours carry over with an offset.
- Rules needing a capability the new model lacks are paused and flagged on the object page.

7.6 Data quality: jumps, drift and late data

Problem	Handling
Invalid fix, 0,0, too few satellites, poor accuracy	Flagged invalid; not shown or counted
Impossible jump	Implied speed between points above the object-type limit (e.g. car 200 km/h, pet 50 km/h) → flagged
Parked drift	With ignition off and no motion, position frozen; movement under ~30 m ignored
Duplicates / future timestamps	Dropped / clamped and flagged
Late and out-of-order points	Inserted by fix time; trips and daily summaries for that period recalculated
LBS / Wi-Fi fixes	Shown as approximate; excluded from distance

> Flagged points are kept (never deleted) and raw packets are logged, so data can always be audited or reprocessed with better rules.

7.7 Commands

Commands go out through the device's open TCP connection when online; otherwise they wait in a queue with an expiry and can fall back to SMS. Each command records who sent it, when, the device response and final status.

8 Data model

Main entities, grouped by domain. Every tenant-owned table carries `company_id` (and `partner_id` through the company) for isolation. Times are stored in UTC.

Domain	Entity	Key fields / relations
Tenancy	Partner	name, branding, domain, wholesale pricing
Tenancy	Company	partner, account type (Business/Individual), status, timezone, units, retention days, modules, plan
Tenancy	Branch	company, name, address
Access	User	company, name, email, mobile, password hash, 2FA, status, manager (parent user)
Access	Role, Permission, MenuPermission	role → menu → actions (V/C/E/D/special)
Access	UserRole, UserBranch, UserObject	user ↔ role, user ↔ branches, user ↔ individual objects
Access	Session, OtpCode, PasswordReset, ApiKey, AuditLog	security and traceability
Catalogue	DeviceModel	brand, model, protocol, capabilities, field definitions, commands, time offset, filter defaults
Assets	Device	IMEI/unique ID, model, SIM, company, status, firmware, ingest token
Assets	Sim	number, operator, expiry, device
Assets	TrackedObject	company, branch, type, name, icon, type fields, custom fields, speed limit, odometer offset
Assets	DeviceAssignment	device ↔ object, from, to
Assets	VehicleGroup (+ members)	saved groups for filters/reports
Telemetry	Position *(hypertable)*	object, device, fix time, lat, lng, speed, course, altitude, satellites, accuracy, ignition, flags, attributes (JSON)
Telemetry	RawPacket *(hypertable, short retention)*	device, received at, bytes
Telemetry	LiveState *(Valkey)*	last valid position, status, status since, geofence membership, trip state
Telemetry	DailySummary	object, local date, distance, running/idle/stop/offline seconds, max/avg speed, trips, fuel
Telemetry	AutoTrip, Stop	object, start/end, distance, duration, speeds
Geo	Geofence (+ versions)	company, branch, shape, geometry, category, colour, rules, time windows, applies-to
Geo	Poi	name, location, radius, category
Alerts	AlertRule	scope, type, parameters, schedule, cooldown, channels, recipients, escalation

Domain	Entity	Key fields / relations
Alerts	AlertSetting	level (company/branch/group/object), target, alert type, overridden fields only, priority among groups
Alerts	AlertTemplate	named pack of alert settings applied to groups/objects
Alerts	AlertSnooze	object or group, alert types, until, reason, by
Alerts	UserAlertSubscription, ContactGroup	per-user alert types/objects/channels; external contacts for notifications
Alerts	MessageTemplate (+ versions)	level (system/partner/company), purpose (alert type or system message), channel, language, subject, body, placeholders, DLT template ID, sender ID, approval status
Alerts	RecipientList	level, target, alert type, severity, channel, mode (default/add/only/none), entries (user, role, dynamic, contact group, email, mobile)
Alerts	Alert	rule, object, type, priority, time, location, data, status, assignee, notes, ack/resolve by and when
Alerts	NotificationLog	alert, channel, recipient, status, cost
Commands	Command	device, type, payload, sent by, channel, status, response, expiry
Drivers	Driver	personal, licence, badge, medical, police verification, RFID, status
Drivers	DriverAssignment	driver ↔ object, from, to
Routes	Route, RouteStop	ordered stops, planned times, path, corridor, schedule
Routes	Trip, TripStop	route snapshot, driver, vehicle, status, planned vs actual per stop
Passengers	Student, Employee, Guardian	profiles; guardian ↔ student
Passengers	PassengerAssignment	passenger ↔ vehicle, route, pickup/drop stop, shift, effective from/to
Passengers	Attendance	trip, passenger, boarded/dropped time and location, source (RFID/app)
Fuel	FuelTank, FuelCalibration, FuelEvent, FuelEntry	tanks, calibration tables, refuel/theft events, manual fill-ups
Maintenance	ServiceSchedule, ServiceRecord, Document	km/date/hours schedules, logs, expiring documents
Video	VideoClip, VideoSession	device, channel, time range, storage key / who watched
Reports	ReportSchedule, ReportRun	owner, scope, types, frequency, format, recipients / generated files
Billing	Plan, Subscription, Invoice, Payment, CreditPack	limits, prices, Razorpay IDs, GST details
Integrations	Webhook, ForwardTarget	URL, events, secret, status

8.1 Storage estimates

Assumption	2,000 devices	10,000 devices
Average positions per device per day (mix of moving and parked)	~2,900	~2,900
Positions per day	~5.8 million	~29 million
Peak ingestion (10 s interval, all moving)	~200 / s	~1,000 / s

Assumption	2,000 devices	10,000 devices
Compressed position storage, 180 days	~25–40 GB	~130–200 GB
Raw packet log (30 days)	~30 GB	~150 GB

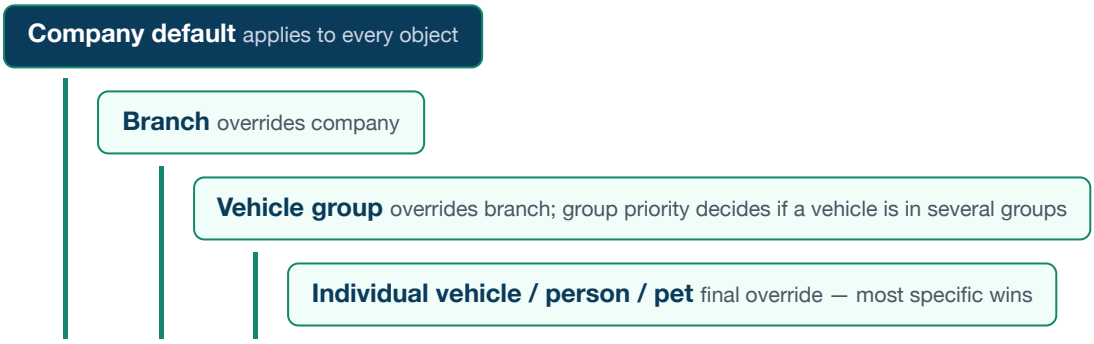
Video storage is sized separately (about 0.5 GB per stream-hour) and governed by video retention per plan.

9 Alerts engine

9.1 Rule structure

Part	Description	Example
Scope	Objects, branch, vehicle group, device model, or all	All trucks in Chennai branch
Type + parameters	From the alert catalogue, filtered by capability	Geofence exit — "Chennai Depot"
Schedule	Days and time windows in company timezone	Mon–Sat, 20:00–06:00
Cooldown / de-duplication	Minimum gap between repeats per object	15 minutes
Priority	Info, Warning, Critical (critical stays on screen)	Critical
Channels & recipients	In-app, push, email, SMS, WhatsApp, webhook — recipients per channel	SMS to owner, email to manager
Escalation	If not acknowledged in X min, notify next level	After 10 min → Admin

9.2 Configuration levels and inheritance



Example: Overspeed	Setting	Effective for
Company default	On, 80 km/h, push + in-app	All vehicles
Branch: Chennai	Adds SMS to branch manager	Chennai vehicles
Group: School buses	40 km/h, priority Critical, 06:00–19:00	School buses (any branch)
Vehicle: TN-01-AB-1234	Off until 25 Oct (in workshop)	That vehicle only

- Resolution is per alert type and per setting, so a vehicle can inherit the schedule from its group but have its own threshold.
- The resolved configuration per object is pre-computed and cached in the rules engine; any change at any level is recompiled for affected objects within seconds, with no impact on alert latency.
- Recipients are resolved at send time: level recipients + user subscriptions + contact groups, filtered by each user's data scope (nobody receives alerts for vehicles they can't see).

9.3 Overspeed and long-idling settings

Every parameter below can be set at company, branch, vehicle group or individual level. Defaults are starting values only.

Overspeed

Setting	Meaning	Default
Speed limit	Limit for the level; vehicle-type defaults offered (bike, car, truck, bus, school bus)	Car 80, truck 60, bus 60, school bus 40, bike 60 km/h
Tolerance	Allowance above the limit before it counts	+5 km/h
Minimum duration	Must stay over the limit this long (filters GPS spikes)	10 s
Severity tiers	Warning when over the limit; Critical when over by X	Critical at +20 km/h
Time-based limit	Different limit in a time window	Off (e.g. night 22:00–05:00 → 50 km/h)
Zone limit	Lower limit inside chosen geofences (school, plant, city)	Set per geofence
Repeat	One alert per overspeed event; optionally repeat every N min while it continues	No repeat
End-of-event summary	When speed drops below limit: max speed, duration, distance over limit	On (in-app only)
Schedule, channels, recipients, cooldown	As for every alert	Inherited

> Overspeed is calculated on the server from clean GPS points, so it behaves the same for every device brand. Where a device has its own overspeed alarm, the limit can also be pushed to the device by command. Each event feeds the driver score and overspeed report.

Long idling

Setting	Meaning	Default
Idle definition	Engine on and speed below threshold	Speed < 3 km/h
Engine-on source	Ignition input; for devices without it, external voltage above a level or OBD RPM > 0	Ignition, else voltage > 13.2 V
Idle threshold	Minutes of continuous idling before alerting	10 min
Severity tiers	Warning at the threshold, Critical at a longer duration	Critical at 30 min
Movement tolerance	Small creeping movement that does not reset the idle timer	< 50 m
Repeat	Re-alert every N min while idling continues	Every 15 min
Exclusions	No alert inside chosen geofences (loading bay, depot) or time windows	None
AC idling	Separate alert when idling with AC on (devices with AC input)	Off
End-of-idle summary	Total idle duration and estimated fuel wasted (fuel sensor, or idle L/h per vehicle type)	On (in-app only)

Setting	Meaning	Default
Schedule, channels, recipients, cooldown	As for every alert	Inherited

9.4 Alert recipients: email, SMS, WhatsApp and push lists

Who receives each alert is configured per alert type, per channel, at any level (company, branch, group, vehicle).

Company default lists (Settings → Notifications)

List	Example
Default email list	ops@abclogistics.in, owner@abclogistics.in
Default SMS numbers	+91 98400 12345, +91 94440 67890
Default WhatsApp numbers	+91 98400 12345
Default push users	All Admins

Per-alert recipient settings

Channel	Mode	Recipients (example: Fuel theft, Chennai branch)
Email	Default + additional	Default list + accounts@abclogistics.in
SMS	Only this list	+91 98400 12345 (owner), Branch Manager (dynamic)
WhatsApp	Use default list	—
Push	Default + additional	Default push users + current driver's supervisor

- **Modes per channel:** Use default list · Default + additional · Only this list · None.
- **Recipient types:** named users, roles (e.g. all Branch Managers), dynamic contacts (users assigned to the vehicle, current driver, branch manager), contact groups, or typed emails/mobile numbers for people without a login.
- **Severity routing:** each severity tier can use different channels, e.g. Warning → email only; Critical → email + SMS + WhatsApp.
- **Inheritance:** lists follow the same Company → Branch → Group → Vehicle inheritance as other alert settings.
- **Scope safety:** users and roles only receive alerts for objects they are allowed to see; typed external contacts receive exactly what the admin configured.
- **Validation and cost control:** email and +91 mobile validation, duplicates removed, optional OTP verification for external numbers, SMS/WhatsApp usage estimate and plan limits shown before saving.
- **Send test** checks every list; the delivery log shows sent / delivered / failed per recipient.

9.5 Message templates

All notification text is template-driven and can be viewed and edited in **Settings → Message templates** (Super Admin for system defaults, company Admin for the company's own versions).

Channel	What can be edited	Approval needed
Email	Subject and full body (rich text/HTML), logo, colours, footer, placeholders	None — changes apply immediately
Push	Title, body, sound, priority	None

Channel	What can be edited	Approval needed
SMS	Choice of template per alert, variable mapping; new wording via request	**Yes — TRAI DLT.** Text must match a DLT-registered template exactly
WhatsApp	Choice of template per alert, variable mapping; new wording via request	**Yes — Meta** template approval

Example: Overspeed

Channel	Template
Email subject	`[{{severity}}] Overspeed: {{vehicle_name}} at {{speed}} km/h`
Email body	`{{vehicle_name}} ({{plate}}) was driving at {{speed}} km/h (limit {{limit}}) near {{location}} at {{time}}. Driver: {{driver_name}}. View on map: {{map_link}}`
SMS (DLT)	`VeeTracer Alert: {#var#} overspeed {#var#} km/h at {#var#} on {#var#}. Track: {#var#}` → mapped to vehicle_name, speed, location, time, map_link
Push	`Overspeed · {{vehicle_name}} / {{speed}} km/h near {{location}}`

Template Master page

Settings → **Message templates** opens the Template Master: a single list of every template the user may see.

Name	Purpose	Channel	Level	Status	DLT / Meta ID	Used by	Last changed
Overspeed – Email	Alert: Overspeed	Email	Company	Active	—	12 rules	Priya, 14 Oct
Overspeed – SMS	Alert: Overspeed	SMS	System	Approved	1107169…	8 rules	Super Admin, 2 Oct
Fuel theft – WhatsApp	Alert: Fuel theft	WhatsApp	System	Pending	fuel_theft_v2	0	Super Admin, 16 Oct
Geofence exit – Push	Alert: Geofence exit	Push	System	Active	—	21 rules	Super Admin, 1 Oct
Login OTP – SMS	System: OTP	SMS	System	Approved	1107169…	All logins	Super Admin, 1 Oct
Monthly report – Email	System: Scheduled report	Email	Company	Draft	—	3 schedules	Ravi, 17 Oct

- **Filters and search:** channel, alert type / system message, level (system/company), status (draft, active, pending approval, approved, rejected, inactive), language, "overridden by my company", "not approved".
- **Row actions:** view, edit, preview, send test, duplicate, version history, reset to default, activate/deactivate.
- **Company Admin view:** system defaults and the company's own overrides, with a badge showing which are overridden.
- **Super Admin view:** all system templates, how many companies override each one (drill down to the list), and every pending DLT/Meta approval.
- **Export** to Excel/CSV, e.g. to prepare the DLT registration batch.

SMS DLT workflow

Draft text → **Register on DLT portal** → **Enter DLT template ID + sender ID** → **Approved — selectable per alert**

- VeeTracer registers a standard set of SMS templates for all alert types before launch, so companies can use SMS from day one.
- A company that wants different SMS wording raises a request from the template screen; once registered on DLT, the template becomes available to them.
- Templates under a white-label partner's own DLT registration are managed by the partner (P4).

9.6 Evaluation

- Rules are compiled per object and held in memory by the rules engine; changes are pushed instantly.
- Previous state (inside/outside each geofence, over speed limit, ignition, status) lives in memory and Valkey, so each check is a constant-time comparison.
- Time-based alerts (offline, idling too long, stayed too long, trip not started, late at stop, document expiry) run on a scheduler.
- Anti-flapping on geofences and status changes: minimum dwell and boundary buffer.
- Notification sending is asynchronous with retries; failures never delay other alerts.
- SMS/WhatsApp usage is metered per company against plan allowance and credit packs.

10 Master pages

Every master in VeeTracer has its own master page built on one standard pattern, so all lists look and behave the same on web (and in a compact form on mobile). Visibility and actions follow the user's role permissions and data scope.

10.1 Standard behaviour of every master page

ID	Feature	Phase
MST-01	List view with search, filters, sorting and pagination (server-side, fast at 10,000+ rows)	P1
MST-02	Column chooser and saved filter views per user	P1
MST-03	Add, view, edit, delete — buttons shown only if the role has that permission	P1
MST-04	Active / Inactive status instead of hard delete where history depends on the record	P1
MST-05	Delete protection: records in use (e.g. a route with trips) can only be deactivated	P1
MST-06	Bulk actions: activate, deactivate, delete, assign branch/group	P1
MST-07	Excel/CSV import with template download, row-level validation and error report	P2
MST-08	Excel/CSV/PDF export of the current filtered list	P1
MST-09	Record history: who created/changed what and when (from the audit log)	P1
MST-10	Custom fields shown as extra columns and filters where the company has defined them	P1
MST-11	Detail page with related records (e.g. a driver's vehicles, trips, alerts)	P1
MST-12	Data scope respected: Managers and Users only see records in their branches/objects	P1

10.2 Master catalogue

Area	Master page	Managed by	Phase
Platform	Companies (customers)	Super Admin	P1
Platform	Partners (white-label)	Super Admin	P4
Platform	Plans & pricing, Coupons	Super Admin	P1 / P2
Platform	Device models & capabilities	Super Admin	P1
Platform	Device inventory, SIM inventory	Super Admin	P1
Platform	Command catalogue	Super Admin	P1
Platform	Message templates (Template Master)	Super Admin / Company Admin	P1
Platform	Alert types catalogue	Super Admin	P1
Organisation	Branches	Company Admin	P1
Organisation	Users	Admin / Manager	P1
Organisation	Roles & menu permissions	Company Admin	P1
Organisation	Contact groups	Admin / Manager	P2

Area	Master page	Managed by	Phase
Organisation	Custom fields	Company Admin	P1
Assets	Vehicles / tracked objects	Admin / Manager	P1
Assets	Devices (company view), SIMs	Admin	P1
Assets	Vehicle groups	Admin / Manager	P1
Drivers	Drivers (with licence, badge, documents)	Admin / Manager	P1
Routes & trips	Routes, Stops / Points of interest	Admin / Manager	P1
Routes & trips	Trip schedules	Admin / Manager	P1
Geofences	Geofences	Admin / Manager	P1
Alerts	Alert settings (by level), Alert templates (packs), Snoozes	Admin / Manager	P1
Alerts	Recipient lists, Default recipient lists	Admin	P1
Passengers	Students, Guardians, Employees	Admin / Manager	P2
Passengers	Shifts	Admin	P2
Fuel	Fuel tanks & calibration tables	Admin	P2
Maintenance	Service schedules, Vehicle documents	Admin / Manager	P2
Reports	Report schedules	Any user (own) / Admin (all)	P1
Integrations	API keys, Webhooks, Forwarding targets	Admin	P1 / P2

10.3 Lookup masters (dropdown values)

Small reference lists used in dropdowns. VeeTracer ships system values (Super Admin); companies can add their own where marked.

Lookup	Examples	Company can add
Object types & vehicle types	Car, Bike, Truck, Bus, School bus, Van, Person, Pet, Asset	Vehicle sub-types
Vehicle makes & models	Tata, Ashok Leyland, Mahindra, Maruti, Hero, TVS…	Yes
Fuel types	Petrol, Diesel, CNG, EV	No
Licence classes	MCWG, LMV, LMV-TR, HMV, HGMV, HPMV, Transport	No
Document types	Insurance, Permit, Fitness, PUC, RC, Licence, Badge	Yes
Geofence / POI categories	Home, Office, Depot, Customer, School, Parking, Fuel station, No-go	Yes
Alert priorities	Info, Warning, Critical	No
Departments, designations	—	Yes
Classes / sections, campuses	—	Yes
Timezones, units, languages	Asia/Kolkata, km / mi, English	No

11 Apps, roles and animations

11.1 Which app shows what

Role	App	Home screen	Main menus
Super Admin	Web, Admin app	Platform dashboard	Customers, devices, plans, catalogue, templates, health
Company Admin / Manager	Web, Admin app	Fleet dashboard	Live map, vehicles, drivers, routes & trips, geofences, alerts, reports, masters, users
User / Viewer	Web, VeeTracer app	Fleet dashboard (own scope)	Live map, vehicles, history, alerts, reports (as permitted)
Individual	Web, VeeTracer app	My Vehicles	Live map, history, my rides, alerts, geofences, family
Driver	VeeTracer app	Today's duty & trip	Trip, stops, SOS, inspection, fuel entry, my score, documents
Attendant	VeeTracer app	Today's trip passengers	Passenger list, attendance, SOS
Employee (passenger)	VeeTracer app	My cab today	Track my cab, ETA, boarding, SOS, roster, history
Delivery staff	VeeTracer app	Duty on/off	Duty, my location, SOS
Parent	Parent app	My children	Bus live, ETA, boarding status, trip history

Menus are never hard-coded per app: each app requests the user's permissions at login and builds its menu from the menu-role mapping, so a new custom role works in the apps without an app update.

11.2 Design system and themes

ID	Feature	Phase
UI-01	Web UI built with Tailwind CSS; accessible components from shadcn/ui (Radix), owned in our codebase	P1
UI-02	Light, Dark and System (follow device) themes on web and all mobile apps	P1
UI-03	Theme choice saved per user and synced across web and mobile	P1
UI-04	Shared design tokens (colours, spacing, radius, typography) used by Tailwind config and Flutter ThemeData, so web and apps look the same	P1
UI-05	Status colours (running, idle, off, overspeed, offline) and alert priorities tuned for contrast in both themes	P1
UI-06	Light and dark map styles from our own OSM tiles; map switches with the app theme (or user can lock map to light)	P1
UI-07	Charts, tables, dialogs and playback controls fully themed	P1
UI-08	Responsive layouts: desktop, tablet and mobile web	P1
UI-09	Accessibility: WCAG AA contrast in both themes, keyboard navigation, visible focus, 'reduce motion'	P1
UI-10	White-label partners override brand tokens (logo, primary colour) without code changes	P4

Item	Light	Dark
Web app, Admin app, VeeTracer app, Parent app	Yes	Yes
Map tiles	Light style	Dark style
Emails	Light (email clients handle dark mode inconsistently)	Auto-adapted by the email client
PDF / Excel reports, invoices	Always light (print-friendly)	—

11.3 Animations and trip playback

ID	Feature	Phase
ANI-01	Trip playback animation everywhere a trip appears (history, trip details, driver/employee/parent trip history, alerts, reports)	P1
ANI-02	Playback controls: play/pause, 1×–16× speed, scrubber, jump to stop/alert, follow-camera on/off	P1
ANI-03	Path drawn progressively behind the moving marker, coloured by speed or status (running/idle/overspeed)	P1
ANI-04	Marker rotates smoothly with heading; vehicle-type icon (car, bike, bus, truck, person, pet)	P1
ANI-05	Live data panel during playback: time, speed, ignition, AC, fuel, address, driver; speed graph cursor in sync	P1
ANI-06	Stops and alerts pop in on the map as playback passes them	P1
ANI-07	Planned route shown as dashed line with actual path animating over it (route trips)	P1
ANI-08	Live map: markers glide smoothly between updates (interpolated position and bearing) instead of jumping	P1
ANI-09	Follow mode with smooth camera pan and auto-zoom by speed	P1
ANI-10	Trip cards with a 'mini replay' preview animation on tap	P2
ANI-11	Parent/employee view: bus/cab moving along the route line towards 'my stop' with animated ETA progress	P2
ANI-12	Geofence entry/exit ripple, stop-arrival pulse, new-alert slide-in with sound	P1
ANI-13	Status colour transitions, animated dashboard counters and charts, skeleton loaders, smooth page transitions	P1
ANI-14	Export trip replay as a shareable video/GIF	P4
ANI-15	Performance guard: GPU map rendering (MapLibre); marker animation only for objects in view and below a count limit; 'reduce motion' and battery-saver settings respected	P1

> Trip playback runs entirely on the device from the downloaded track (simplified for long trips), so it stays smooth at 60 fps and puts no load on the server while playing.

12 Live map, history and trips

12.1 Live map

- On open, the client loads the current state of every object in scope from Valkey (fast snapshot), then receives only changes.
- The server knows each viewer's scope and current map area; updates outside them aren't sent.
- Updates are batched about once per second per viewer, and markers are animated between points.
- At 10,000 objects the web map uses clustering and WebGL rendering (MapLibre) to stay smooth.

12.2 Automatic trips vs planned trips

	Automatic trip	Planned (route) trip
Created by	System, from ignition/movement	User/schedule: Driver + Vehicle + Route + time
Who uses it	Everyone, including Individuals	Business accounts with routes
Ends when	Stop longer than 5 min (configurable)	Arrival at final stop or manual close
Extra data	—	Planned vs actual per stop, delays, passengers, on-time %

12.3 Planned trip states

Scheduled → **Started** → **In progress** → **Completed** / **Incomplete** / **Cancelled**

13 Reports

Report	Contents	Phase
Summary (day / week / month)	Distance, running, idle, stop, offline time, trips, max/avg speed, overspeed count, first start, last stop, engine hours, fuel	P1
Trips	Automatic and planned trips with details	P1
Stops / idle	Location, start, duration	P1
Alerts	All alerts with acknowledgement details	P1
Trip performance	On-time %, stop delays, planned vs actual km by route/driver/vehicle	P1
Geofence visits	In, out, duration, violations	P2
Ignition / AC / engine hours	On/off times and durations	P2
Overspeed	Events with peak speed, duration, location	P2
Utilisation	Running vs available time per vehicle	P2
Fuel	Consumption, km/L, refuels, theft	P2
Driver score	Score, events, ranking	P2
Attendance	Boarding/drop logs, absentees, occupancy	P2

Daily summaries are maintained continuously per object, so weekly and monthly reports for a whole fleet are sums over stored daily rows rather than scans of raw positions.

14 Passenger module and Parent app

- Students and employees are mapped to a vehicle, route, pickup stop and drop stop (separately for morning and evening).
- Each planned trip carries the passenger list per stop.
- Attendance per company: Off, RFID, in-app, or both. RFID taps and attendant marks merge into one record.
- Parent app login by mobile OTP, matched to the guardian phone in the student profile.
- Parents see only their own children: profile, assigned bus, live location during trip hours, ETA, boarded/dropped status, trip history.
- Parent and employee notifications: bus started, approaching stop, boarded, dropped, delayed.
- Built behind a module switch so it can be included in plans or sold as an add-on later.

15 Video and cameras

- Devices stream to ZLMediaKit, which converts JT/T 1078, RTMP and RTSP into WebRTC (lowest delay), HTTP-FLV or HLS for web and mobile.
- Streams start only when someone opens them and stop automatically after a set time.
- Alarm clips and snapshots are stored in object storage, linked to alerts and the history timeline.
- Separate permissions for live view, playback and download; every view is logged.
- Each live stream needs ~0.5–1 Mbps on the server; 50 concurrent viewers ≈ 50 Mbps upload. The media server scales independently.

> Open item: the exact Concox/Jimi dashcam model must be confirmed. Some firmware uses JT808/1078, others only Jimi's own platform.

16 Maps (self-hosted OpenStreetMap)

Service	Tool	Resources (India)
Vector tiles	Planetiler (build) + Martin (serve), MapLibre (display)	~10 GB tiles, light CPU
Reverse geocoding	Nominatim (or Photon)	~16–32 GB RAM, ~150–300 GB SSD
Routing / ETA / snapping	OSRM	~8–16 GB RAM

- Source data: India extract from Geofabrik (~1.5 GB), refreshed monthly.
- Address lookups cached by rounded coordinates to keep load low.
- No per-request map fees. Satellite imagery is the only map feature that would need a paid provider.

17 Billing and plans

17.1 Structure

- Separate plan families for Business and Individual accounts.
- Each plan has a monthly price and an annual price (proposed: annual = 10 × monthly, i.e. 2 months free).
- Limits per plan: devices, users, history days (≤ 180), geofences, SMS/WhatsApp allowance, video minutes, modules.
- Razorpay Subscriptions for auto-renewal; grace period then automatic suspension; reactivation on payment.
- GST-compliant invoices with CGST/SGST or IGST and customer GSTIN.

17.2 Illustrative plan grid (prices to be decided)

Plan	Account	Devices	Users	History	Channels	Monthly / Annual
Personal	Individual	1–2	3 (family)	30 days	In-app, push, email	₹ TBD
Family	Individual	up to 5	5	60 days	+ SMS allowance	₹ TBD
Starter	Business	up to 10	5	60 days	In-app, push, email	₹ TBD
Business	Business	up to 100	25	90 days	+ SMS, WhatsApp	₹ TBD
Enterprise	Business	custom	custom	180 days	All + webhooks, API	₹ TBD

18 Security, compliance and India-specific requirements

Topic	What VeeTracer does
DPDP Act 2023	Consent capture for tracking people (delivery staff, employees, students via school), purpose limitation, data export/deletion on request, privacy policy
Aadhaar	Never store full number in clear; mask to last 4 digits; encrypted document storage
AIS-140	Certified device support, emergency alert handling, data forwarding to state backends; state-by-state requirements to be confirmed
SMS (TRAI DLT)	Register entity, sender IDs and templates before launch (takes weeks)
WhatsApp	Meta-approved templates via Cloud API or a BSP
GST	Invoices with GSTIN, place of supply, CGST/SGST/IGST
Application security	OWASP practices, parameterised queries, rate limiting, 2FA, audit log, secrets management, encrypted backups
Data residency	Hosting in India recommended for customer trust and regulatory comfort

19 Infrastructure and running costs

19.1 Recommended servers

Role	Launch (≤ 3,000 devices)	Growth (~10,000 devices)
Application (gateways, API, realtime, workers, Valkey)	1 × 8 vCPU, 16–32 GB RAM	2–3 × 8 vCPU, 16 GB RAM
Database (PostgreSQL + TimescaleDB)	1 × 8 vCPU, 32 GB RAM, 1 TB NVMe	1 × 16 vCPU, 64 GB RAM, 2 TB NVMe + 1 replica
Maps (tiles, Nominatim, OSRM)	Shared with app server or 1 × 4 vCPU, 32 GB	1 × 8 vCPU, 32 GB RAM, 500 GB SSD
Media server (P3)	—	1 × 8 vCPU, 16 GB RAM, 1 Gbps
Object storage	500 GB	4–10 TB (video dependent)
Backups	Off-site daily	Off-site daily + point-in-time recovery

19.2 Running costs that can't be avoided

Item	Indicative cost (verify current rates)
Servers	Depends on provider (own hardware, Indian data centre, or cloud)
Domain + TLS	Domain ~₹1,000/year; TLS free (Let's Encrypt)
SMS	~₹0.15–0.30 per SMS + DLT registration
WhatsApp	Per-message template pricing set by Meta (utility messages are the cheapest category)
Email	~$0.10 per 1,000 (Amazon SES)
Payments	Razorpay ~2% per transaction
App stores	Google Play $25 one-time; Apple $99/year
Cell-tower location	OpenCelliD free; paid services optional for better coverage

20 Non-functional requirements

Requirement	Target
Live update latency (device → screen)	< 2 s at 95th percentile
Alert latency	In-app < 2 s, push < 5 s, SMS/WhatsApp/email < 30 s
Ingestion capacity	1,000 positions/s sustained at 10,000 devices; burst 3× without loss
Data durability	No accepted position lost (durable stream + acknowledged writes)
Availability	99.9% monthly for API and live tracking
Backups	RPO ≤ 15 min (WAL archiving), RTO ≤ 4 h
Report generation	Monthly summary for 1,000 vehicles < 10 s
Web map	Smooth with 10,000 markers (clustered)
Mobile tracker battery	< 5% per hour on duty at default interval (target, device dependent)
Retention	Configurable 30–180 days per company; video separately

21 Phased roadmap

Phase	Goal	Main contents
P1 Core platform	Sell to the first customers	Multi-tenant accounts (Business/Individual), roles & branches, single Super Admin console, Concox GT06 + phone tracking, live map with statuses, history & automatic trips, drivers, routes & planned trips, geofences, core alerts (in-app/push/email), commands (engine cut-off), summary reports + scheduling, dashboards, Razorpay plans + GST invoices, self-hosted maps, web, VeeTracer Admin app, role-based VeeTracer app (users, individuals, drivers, delivery staff) with trip animations, REST API, load-tested for 10,000 devices
P2 Operations	Depth for fleets and schools	AIS-140, Teltonika, Traccar gateway (other brands), OBD, watches, pet trackers; SMS/WhatsApp/webhooks; escalation; fuel monitoring; driver score; maintenance & documents; passenger module + Parent app; more reports; OSRM routing; multi-screen tracking; share links
P3 Video	Camera-enabled fleets	JT/T 808/1078 dashcams and MDVR, live video, playback, alarm clips, ADAS/DMS alerts
P4 Growth	Scale the business	White-label partners and partner billing, ad-hoc tracking, satellite layer, Sarathi verification, Tamil/Hindi

> Durations depend on team size and are deliberately not fixed here (own timeline). Each phase ends with a usable, releasable product.

21.1 Delivery plan (AI-assisted build)

Planned with Claude writing most of the code, the product owner testing and deciding daily, and ideally one developer reviewing. Times are calendar estimates; they will be re-confirmed at the end of M1 based on measured progress.

Milestone	Deliverable	Estimate
M1 Foundation (trial)	Accounts, roles, masters framework, Concox GT06 gateway, device simulator, live map on web, light/dark theme	2–3 weeks
M2 P1a — launchable core	History and playback animation, geofences, core alerts, summary reports, Razorpay plans, Admin and role-based apps	+6–8 weeks
M3 Full P1	Drivers → routes → trips, alert levels, recipient lists, templates, scheduled reports, dashboards, driver app, 10k load test, production deployment	+6–8 weeks
P2	Operations (see roadmap)	~3–4 months
P3	Video	~1.5–2 months

Prerequisites

- M1: TimescaleDB and Valkey on the development machine (PostgreSQL 16 already installed); 1–2 Concox test devices with data SIMs.
- M2: a server or tunnel reachable by devices; Android test phone; Razorpay test account.
- M3: Xcode and Apple developer account, Google Play account, DLT-registered SMS templates, production server.

21.2 Phase 1 build order

Step	Deliverable
1	Foundations: repo, CI, Docker, database schema, auth (OTP + email/password), tenancy, roles & scope
2	Device catalogue, devices/objects/assignments, Concox GT06 gateway, phone ingestion, stream + processor
3	Live state, realtime gateway, live map (web), statuses, vehicle info panel
4	History, automatic trips, daily summaries, data-quality filters
5	Geofences + rules engine + notifications (in-app, push, email) + alert panel
6	Drivers, routes, planned trips, trip alerts and reports
7	Reports and scheduling, dashboards, commands
8	Billing (Razorpay, GST invoices), Super Admin console
9	Flutter apps: VeeTracer Admin, role-based VeeTracer app (viewer, individual, driver, tracker mode), trip playback animations
10	Self-hosted maps, 10,000-device load test, security review, production deployment

22 Open decisions

#	Decision	Needed by
1	Exact Concox/Jimi dashcam model(s) and their protocol (JT808/1078 or proprietary)	Before P3
2	AIS-140: which states, and which Concox (or other) models are certified	Before P2
3	Plan prices (monthly/annual) for Business and Individual plans	Before P1 launch
4	Passenger module pricing: included or paid add-on	Before P2 launch
5	Hosting provider / own hardware and data-centre location (India recommended)	During P1
6	SMS provider (DLT registration), WhatsApp provider, email provider	Start DLT early in P1
7	Google Play and Apple developer accounts in the company name	During P1
8	Satellite imagery provider (optional, paid)	P4
9	Legal: terms of service, privacy policy, DPDP consent wording, data processing agreement	Before P1 launch
10	Brand assets: logo, colours, app icons for VeeTracer	During P1

23 Glossary

Term	Meaning
AIS-140	Indian standard for vehicle tracking devices in public/commercial transport
DLT	TRAI's Distributed Ledger registration required for commercial SMS in India
DPDP Act	Digital Personal Data Protection Act, 2023 (India)
GT06	The protocol used by most Concox GPS trackers
JT/T 808 / 1078	Chinese standards for vehicle terminals (tracking/alarms) and video streaming, used by most dashcams and MDVRs
LBS	Location from mobile cell towers (approximate)
MDVR	Mobile digital video recorder with several cameras (buses, trucks)
OBD	On-board diagnostics port; OBD trackers read engine data
OSM	OpenStreetMap, the free, open map data source
Hot path	The processing chain that must be fast: incoming position → live map and alerts
Hypertable	TimescaleDB's automatically partitioned table for time-series data
Capability	A feature a device model supports (ignition, AC, fuel, camera…), driving what the UI and alerts offer